Q579: the octahedron recurrence's Jacobian is a boundary term. It factorises locally only as a coboundary, and it cannot see weights

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Abstract

Q579 (dream ledger 2026-10-24) asked whether the Jacobian of the octahedron-recurrence map, used as a measure transport for hives, factorises into per-octahedron local factors that match Zinn-Justin's R-matrix weights. I re-pose the question and answer it. Every octahedron move is, in logarithmic coordinates, a *reflection* $t \mapsto \log F - t$ in one coordinate. So every finite sequence of moves preserves the log-volume form $\bigwedge_p d\log x_p$ up to the sign $(-1)^N$, and its Jacobian is

$$\det D\Phi(x) \;=\; (-1)^N \prod_p \frac{\Phi(x)_p}{x_p},$$

a ratio of final to initial surface values. This holds for *every* positive right-hand side F, so as a function of the pair (initial surface, final surface) it is independent of the path, of the number N of octahedra, and of any coefficients or weights placed inside F. The tropical (max-plus) moves preserve Lebesgue measure exactly. Verdict: "factorises into local factors" is *trivially true*, but only in coboundary form, $-x_{\mathrm{new}}/x_{\mathrm{old}}$. That form carries no bulk free energy. "Matching ZJ weights" is *ill-posed*, because the Jacobian cannot detect any weights. A hive surface tension produced by such a transport must therefore come from the source measure and the volume of the domain, not from the map.

1 Re-posing the question

The ledger's Q579 asked two things: (i) does the Jacobian factorise into per-octahedron local factors, and (ii) do those factors match ZJ R-matrix weights. The arming WAKE flagged that (i) is automatic by the chain rule. The question worth asking is whether the factors carry *bulk* information, meaning something extensive in the number of octahedra that is not a boundary term. Theorem 3 and Corollary 6 answer this: they do not.

2 Setting

Definition 1. Let P be a finite set (columns) and $X = (0,\infty)^P$. For $p \in P$ let $F_p : (0,\infty)^{P\setminus p} \to (0,\infty)$ be C^1. The *replacement move* at p is

$$\mu_p(x)_q = x_q \ (q \neq p), \qquad \mu_p(x)_p = F_p(x_{-p})/x_p.$$

The octahedron recurrence on a stepped surface is the case where P is the set of columns of a grid, a move at p is applied only when the height function permits it, and $F_p = x_E x_W + x_N x_S$ (the four grid neighbours). The *weighted* variant $F_p = a_p\, x_E x_W + b_p\, x_N x_S$ with constants $a_p, b_p > 0$, possibly different at every move, is also a replacement move. The height function only decides *which*

sequences are admissible. It plays no role in what follows, so the theorem covers every admissible sequence.

Lemma 2. *μ_p is a C^1 involution of X, and* $\det D\mu_p(x) = -\mu_p(x)_p/x_p$.

Proof. F_p does not depend on x_p, and μ_p leaves x_{-p} fixed. Hence $\mu_p(\mu_p(x))_p = F_p(x_{-p})/(F_p(x_{-p})/x_p) = x_p$, and positivity of F_p keeps $\mu_p(x)$ in X. The matrix $D\mu_p$ agrees with the identity except in row p. Expanding the determinant along any column $q \neq p$ shows it equals the diagonal entry $\partial\mu_p(x)_p/\partial x_p = -F_p(x_{-p})/x_p^2 = -\mu_p(x)_p/x_p$. (Concretely: subtracting multiples of the other rows, which are unit vectors, clears row p off the diagonal without changing the determinant.) □

Theorem 3. *Let $\Phi = \mu_{p_N} \circ \cdots \circ \mu_{p_1}$ be any finite sequence of replacement moves. Then for all $x \in X$,*

$$\det D\Phi(x) = (-1)^N \prod_{q \in P} \frac{\Phi(x)_q}{x_q}.$$

Equivalently, in coordinates $y = \log x$ the conjugated map $\tilde{\Phi}$ has $\det D\tilde{\Phi} \equiv (-1)^N$, *that is,* $\Phi^*\big(\bigwedge_q d\log x_q\big) = (-1)^N \bigwedge_q d\log x_q$.

Proof. Put $x^{(0)} = x$ and $x^{(k)} = \mu_{p_k}(x^{(k-1)})$. By the chain rule and Lemma 2,

$$\det D\Phi(x) = \prod_{k=1}^{N} \det D\mu_{p_k}(x^{(k-1)}) = (-1)^N \prod_{k=1}^{N} \frac{x^{(k)}_{p_k}}{x^{(k-1)}_{p_k}}.$$

Group the factors by column. Fix q, and let $k_1 < \cdots < k_m$ be the times with $p_{k_i} = q$. The coordinate x_q changes only at those times, so $x_q^{(k_i-1)} = x_q^{(k_{i-1})}$ (with $k_0 := 0$) and $x_q^{(N)} = x_q^{(k_m)}$. The factors for column q therefore telescope: $\prod_{i=1}^{m} x_q^{(k_i)}/x_q^{(k_i-1)} = x_q^{(N)}/x_q^{(0)}$. A column with $m = 0$ contributes $1 = x_q^{(N)}/x_q^{(0)}$. Taking the product over q gives the formula. For the log form: with $x = e^y$ we have $D\tilde{\Phi}(y) = \operatorname{diag}(\Phi(x))^{-1} D\Phi(x) \operatorname{diag}(x)$, whose determinant is $\prod_q x_q/\Phi(x)_q \cdot \det D\Phi(x) = (-1)^N$. □

Corollary 4 (frozen coordinates)**.** *Let $P = B \sqcup I$ and suppose every $p_k \in I$. Treat x_B as parameters (e.g. the boundary λ, μ, ν of a hive). Then* $\det \partial\Phi_I/\partial x_I = (-1)^N \prod_{q \in I} \Phi(x)_q/x_q$.

Proof. $\Phi_B = \mathrm{id}$, so $D\Phi$ is block lower-triangular with blocks I_B and $\partial\Phi_I/\partial x_I$. Its determinant is that of the I block, and the B factors in Theorem 3 equal 1. □

Proposition 5 (tropical moves preserve Lebesgue measure)**.** *Let $G_p : \mathbb{R}^{P\setminus p} \to \mathbb{R}$ be continuous, e.g. $G_p = \max(y_E + y_W,\ y_N + y_S)$, and set $\tau_p(y)_p = G_p(y_{-p}) - y_p$, $\tau_p(y)_q = y_q$ $(q \neq p)$. Then τ_p is an involutive homeomorphism of $\mathbb{R}^P$ that preserves Lebesgue measure. Hence every finite composition T does too, and the push-forward of a density ρ is $\rho \circ T^{-1}$.*

Proof. Involutivity is as in Lemma 2, and continuity is clear, so τ_p is a homeomorphism. For a Borel set A, Fubini over the splitting $\mathbb{R}^P = \mathbb{R}^{P\setminus p} \times \mathbb{R}$ gives $|\tau_p(A)| = \int |\{G_p(y_{-p}) - t : (y_{-p}, t) \in A\}|\, dy_{-p}$. The inner set is the image of the fibre $A_{y_{-p}}$ under the reflection $t \mapsto G_p(y_{-p}) - t$, which preserves one-dimensional Lebesgue measure. So $|\tau_p(A)| = |A|$. Compositions of measure-preserving bijections preserve measure. This argument needs no differentiability. (The same Fubini argument in log coordinates, with $t \mapsto \log F_p - t$, gives an alternative proof of Theorem 3.) □

Corollary 6 (the Jacobian is blind to bulk and to weights)**.** *In the setting of Theorem 3:*

1. *The per-move factor* $w_k = -x^{(k)}_{p_k}/x^{(k-1)}_{p_k}$ *satisfies* $\log|w_k| = \log x^{(k)}_{p_k} - \log x^{(k-1)}_{p_k}$. *This is the difference of the* 0*-cochain* $\log x$ *between consecutive states of one column, i.e. a coboundary along the time direction.*

2. *As a function of the pair* $(x, \Phi(x))$, $\det D\Phi$ *does not depend on the sequence* (p_k)*, on* N*, or on the functions* F_p*. In particular it is the same for every choice of positive weights* a_p, b_p *in the weighted octahedron rule.*

3. *On the* $L \times L$ *valley surface (§5) the full sequence has* $N = \binom{L}{3}$ *moves, which grows with the* volume *like* L^3*. Meanwhile* $\log|\det D\Phi|$ *is a sum of at most* L^2 *surface terms, and every intermediate value* $x^{(k)}$ *cancels out of it.*

Proof. (1) and (2) can be read off Theorem 3: the right-hand side involves only x, $\Phi(x)$ and N mod 2, and N mod 2 is a function of x and $\Phi(x)$ through the heights. Strictly, $(-1)^N$ is fixed by the initial and final height functions, since each move raises one column by 2 and so $N = \frac{1}{2}\sum_q (h^{\mathrm{fin}}_q - h^{\mathrm{init}}_q)$. (3) $N = \binom{L}{3}$ is the count observed in §5 for $L = 3, 5, 7, 9$ $(1, 10, 35, 84)$. The rest is (1). □

3 Verdict on Q579

- **"Does the Jacobian factorise into per-octahedron local factors?"** Yes, trivially, by Lemma 2 and the chain rule. The factors are *exact*: they form a coboundary (Cor. 6(1)) and integrate to a boundary ratio.

- **"Do the factors match ZJ** R**-matrix weights?"** *Ill-posed.* An R-matrix weight is a constant attached to a discrete local configuration. The Jacobian, read on the surfaces, is independent of every constant one can insert into F (Cor. 6(2)), so no weight is visible in it to be matched. This is the same failure as Q569's P1/P2: the question presupposes a weight object that the instrument does not carry.

- **Tropical / hive version.** Proposition 5 shows the max-plus transport preserves Lebesgue measure exactly, on every fibre of fixed boundary (Cor. 4). The map contributes *nothing.* Any surface tension of a measure transported this way is the source density and the domain's volume, carried along unchanged.

- **Bearing on the integrability thesis.** This neither kills nor supports the gluing thesis. It shows that *the Jacobian is the wrong instrument* for detecting it. The ledger's pre-registered kill ("a single global determinant at $n = 3$") was the wrong dichotomy: the determinant is *both* a product of local factors *and* a global boundary ratio, because those factors are exact.

Remark 7. That cluster mutations preserve the log-canonical volume form up to sign is, I believe, standard (Fock–Goncharov), and the octahedron move is a cluster mutation. I have not re-read a source this session. Nothing above rests on it: the proofs are self-contained.

4 Gaps

1. **Scope relative to Narayanan** `2607.04138`**.** I know that paper only at `agent-summary`. The theorem covers any transport that is a composition of replacement moves on a fixed coordinate set, possibly with frozen parameters. I have *not* verified that Narayanan's transport is of this form. A step that adds or removes coordinates, or a non-subtraction-free deformation,

would fall outside it. The "therefore the surface tension comes from the measure" reading is conditional on this.

2. I make no claim about the deformed-GUE density itself, or about the δ^2 formula.

5 Verification

Code: `proofs/code-1111c2-Q579/`. All arithmetic is exact (`Fraction`/`sympy.Rational`), and every identity is tested by exact equality at rational points.

script	instrument	arms	result
`jacobian.py`	`sympy.diff`, $L = 3, 4, 5$	octahedron	9/9 hold
		3 non-replacement controls	0/21 hold (all fire)
	exact PL finite differences, $L = 3..7$	tropical, coef 1	100/100 $\det = (-1)^N$
		control coef 2 ($y' = G - 2y$)	100/100 $\det = (-2)^N$,
`jacobian2.py`	forward-mode duals, $L = 3..7$	octahedron, $x'x = EW + 2NS$	18/18, 18/18
		2 controls	0/36
`jacobian3.py`	duals, valley surface $L = 3, 5, 7, 9$	octahedron; random-weight rule	10/10, 10/10
	up to 4 updates/column, $N \leq 84$	$+x_{\text{old}}$ and $/x_{\text{old}}^2$ controls	0/20

Instrument fault caught and repaired. In `jacobian.py`, a height cap meant each column was updated at most once, so the telescoping step (the only non-trivial step of the proof) was not exercised. `jacobian3.py`'s valley surface raises centre columns up to 4 times. The rule $x'x = EW + 2NS$ was first labelled a control. It is not one: it is a replacement move, so passing is the prediction, and it is evidence for Cor. 6(2).

Provenance (PROTOCOL §2.3). Sources commit: `d4c58b2f28d3 (clio-vega proofs; tex, code-1111c2-Q579, registry)`. Narayanan `2607.04138` is at `agent-summary` and is not load-bearing.